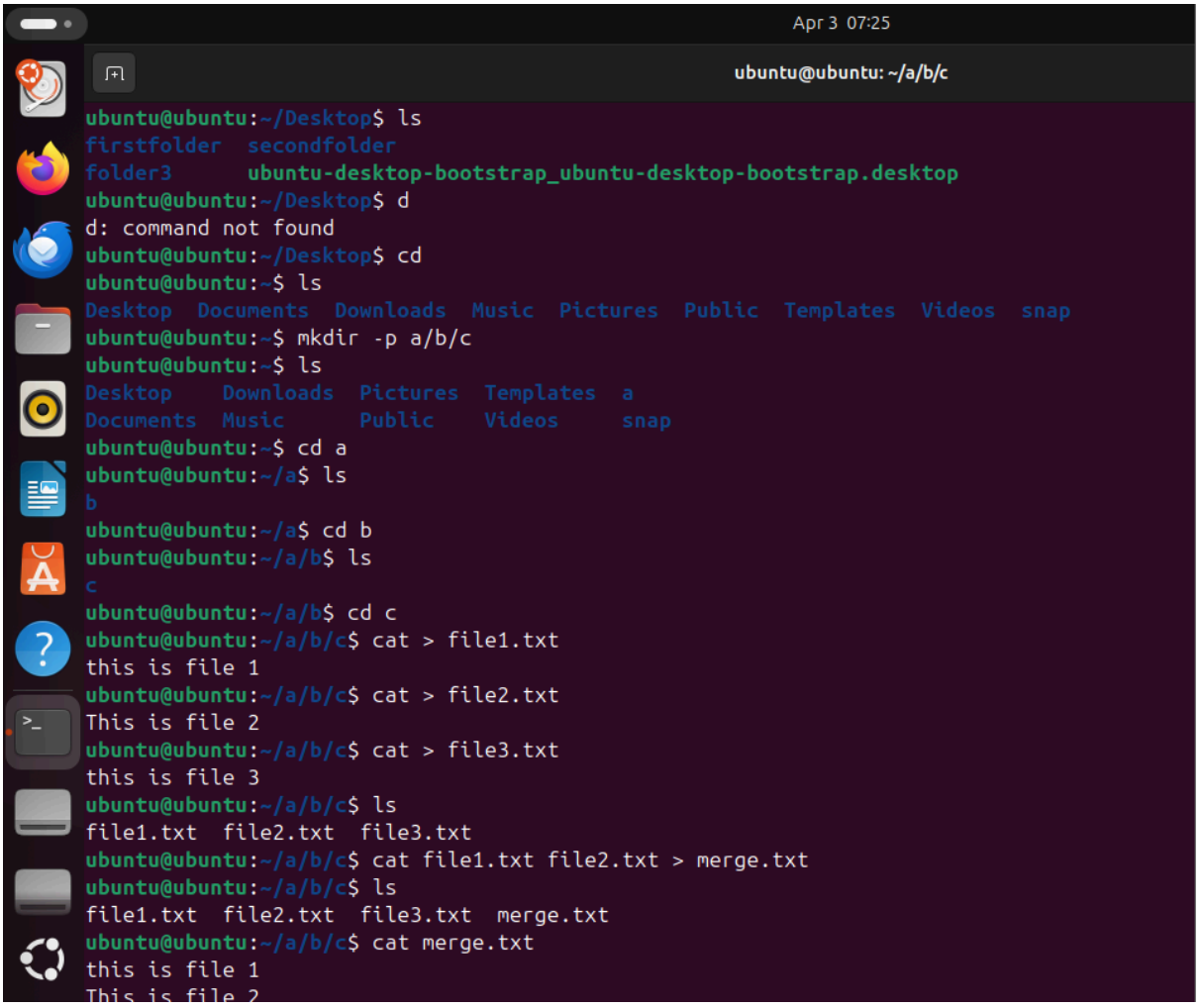
IMPORTANT LAB TASK FROM LAB 01 TO 06

LAB 01:

Lab 1

- **Directory Structure:** Create folders `a`, `b`, and `c` in a parent-child relationship (e.g., `a/b/c`).
- **File Management:** * In folder `c`, create 3 text files.
  - Write content into the first file.
  - Merge the contents of File 1 and File 2.
  - Edit File 2.

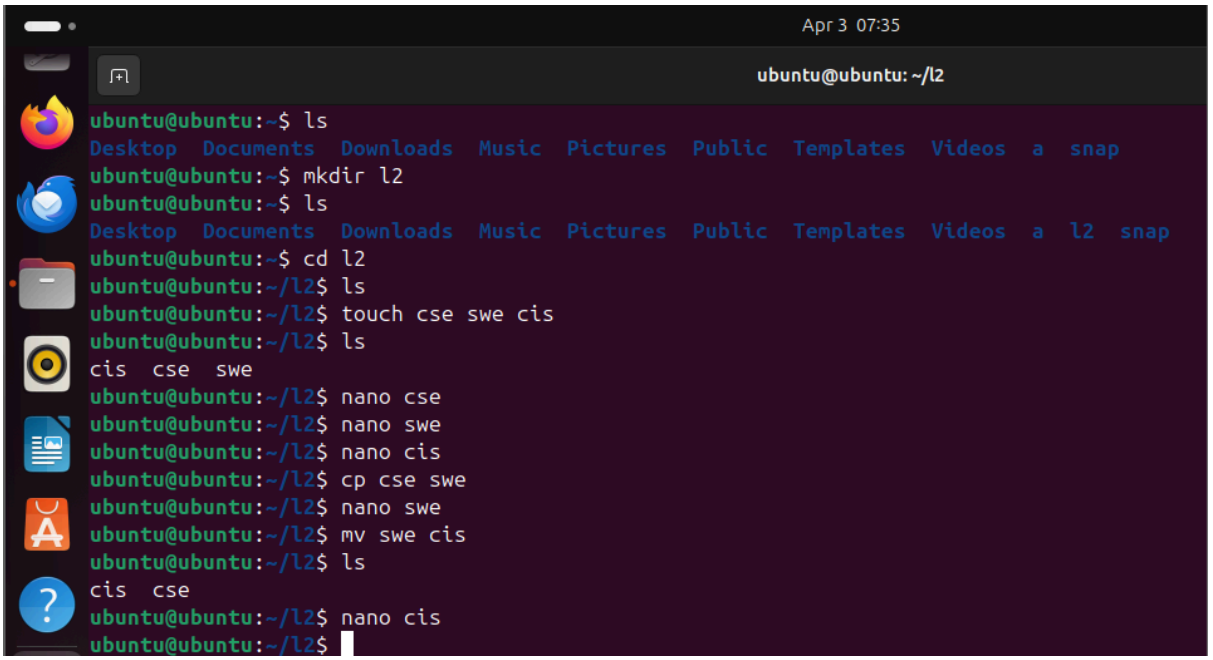


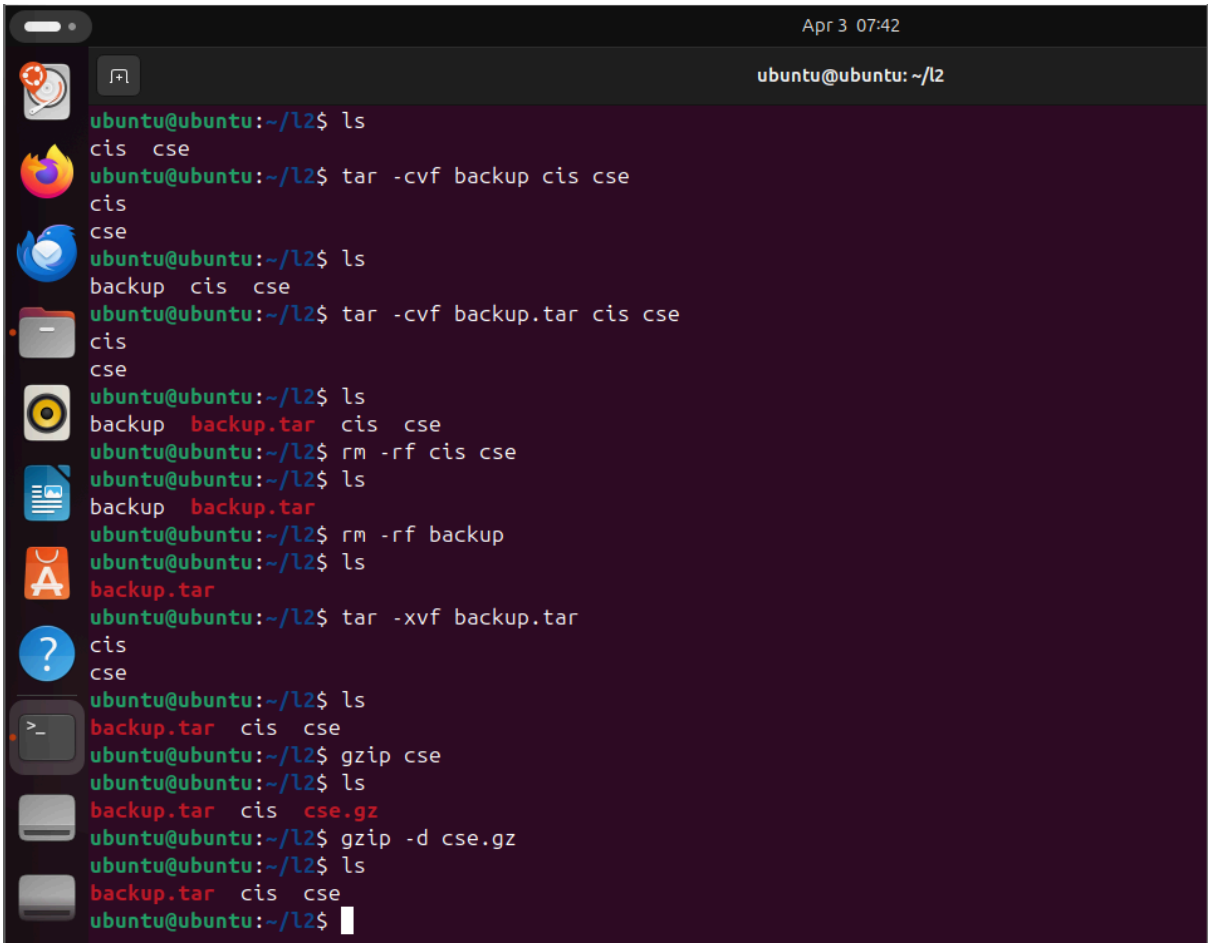
```
ubuntu@ubuntu:~/a/b/c$ cat merge.txt
this is file 1
This is file 2
ubuntu@ubuntu:~/a/b/c$ cat >> merge.txt
This is edit file
ubuntu@ubuntu:~/a/b/c$ cat merge.txt
this is file 1
This is file 2
This is edit file
```

LAB 02:

Lab 2

- **Basic Linux Commands:** * Create 3 files using `touch`: `CSE`, `SWE`, and `CIS`.
  - Write text into each file.
  - Copy `CSE` to `SWE`.
  - Move `SWE` to `CIS`.
- **Search and Compression:**
  - Search for a specific word inside the `CIS` file.
  - Use `tar` to archive `CIS`, then extract the `.tar` file.
  - Use `gzip` to compress `CIS`, then `gunzip` to decompress it.

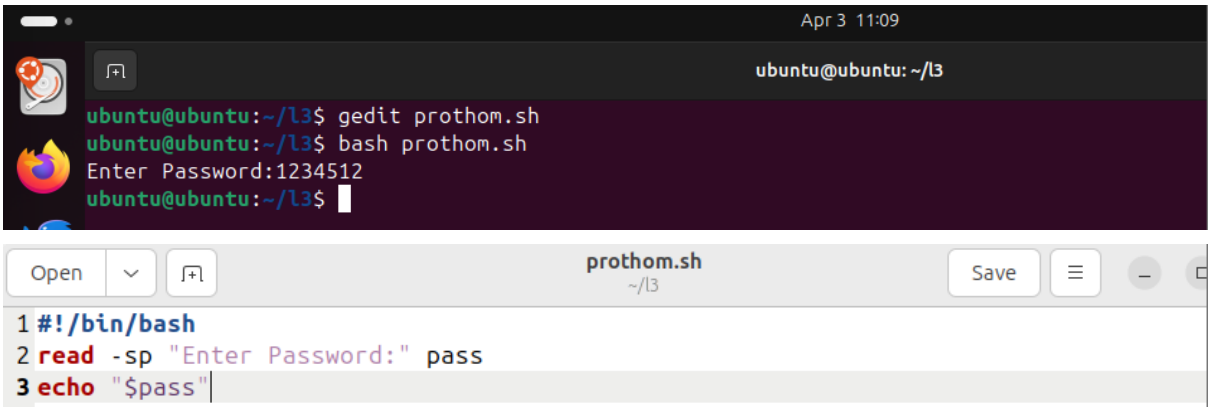


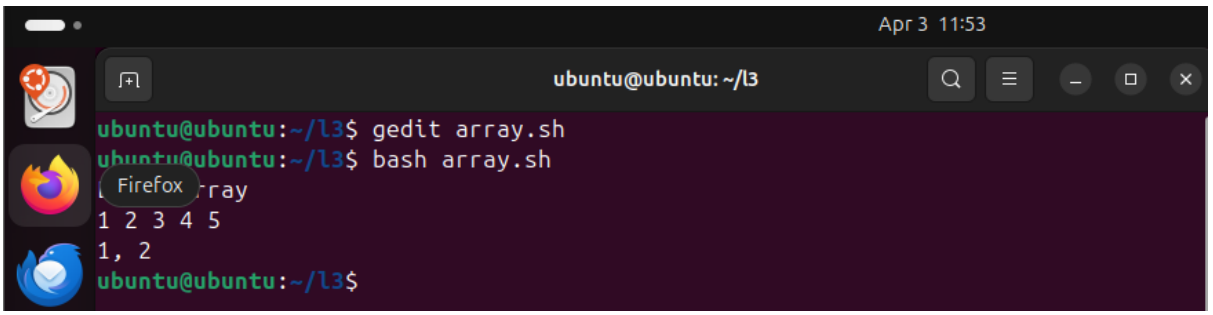


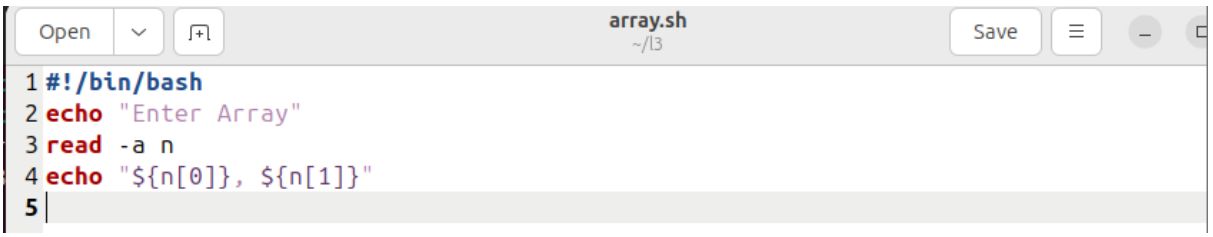
LAB 03:

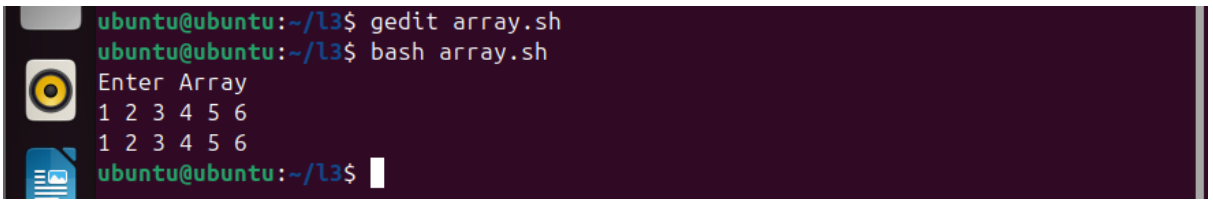
Lab 3

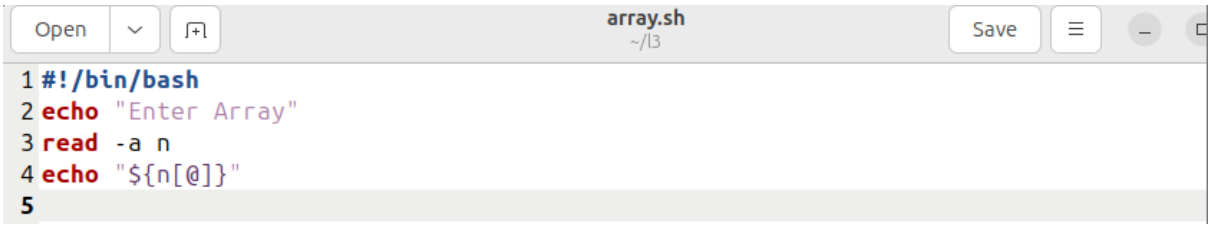
- **Password Masking:** Hide a password based on the
- **Arrays:** Using EOF for input array then print all the array element.







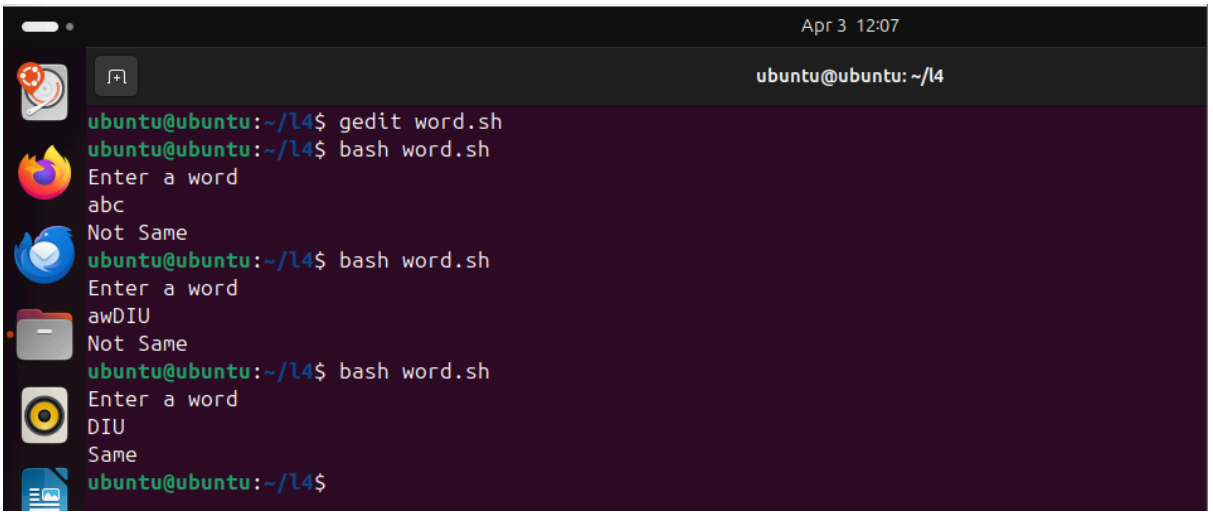




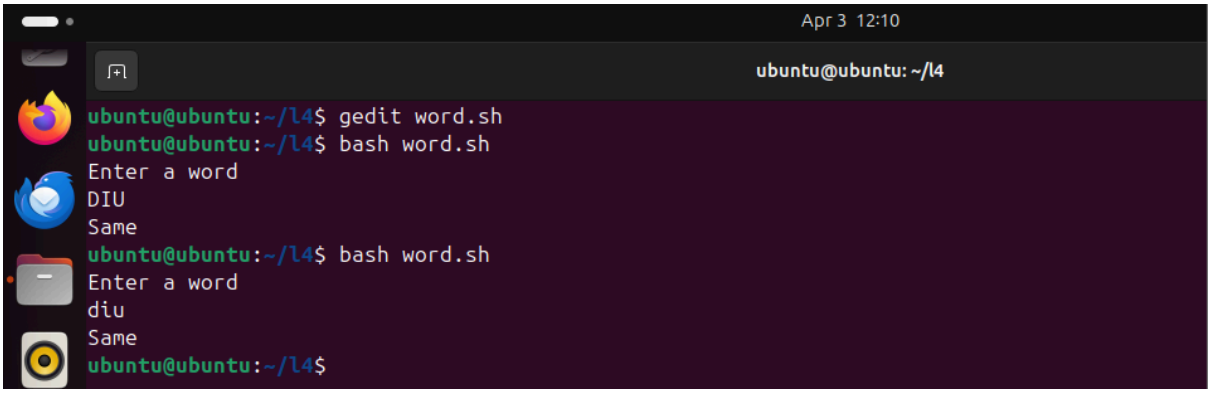
LAB 04:

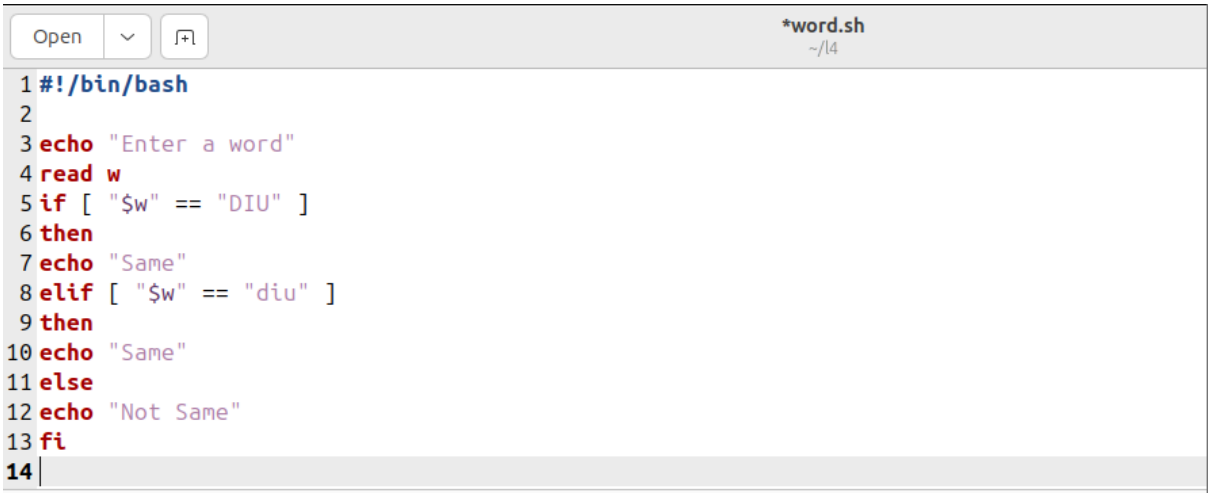
Lab 4

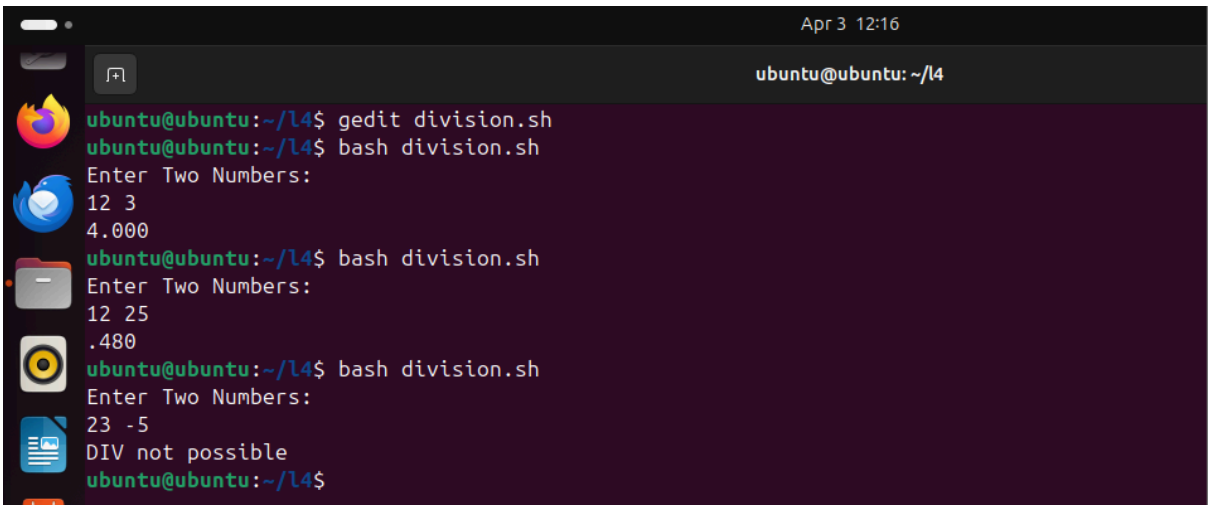
- **String Checking:** Take user input for a word and check if it contains "DIV".
- **Precision and Error Handling:** * Perform division between two numbers and show the output to 3 decimal points.
  - Handle error cases (e.g., division by zero).
- **Grade Generator:** Create a system to generate grades based on marks:

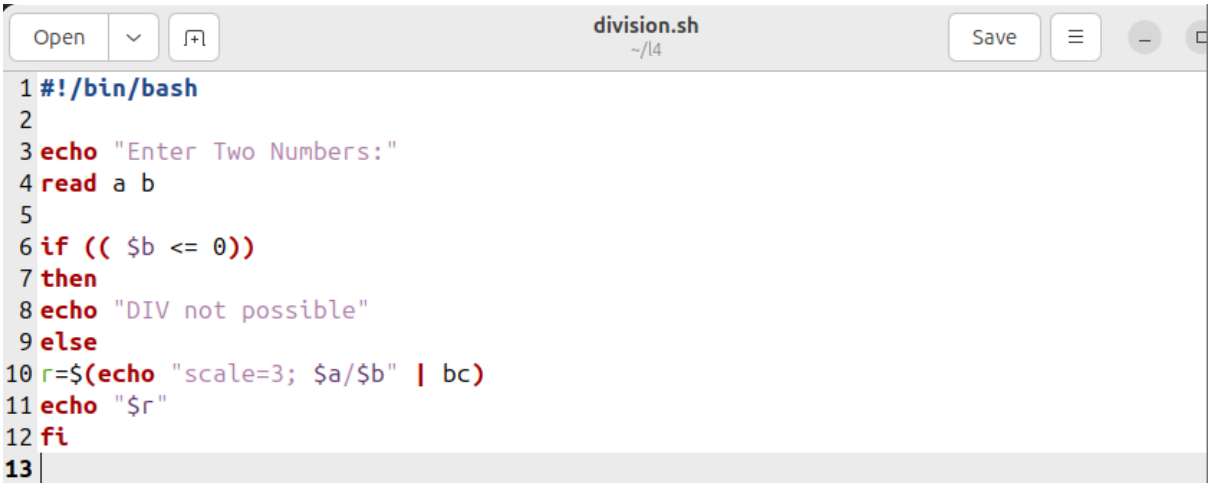

Apr 3 12:07
ubuntu@ubuntu: ~/l4
ubuntu@ubuntu:~/l4$ gedit word.sh
ubuntu@ubuntu:~/l4$ bash word.sh
Enter a word
abc
Not Same
ubuntu@ubuntu:~/l4$ bash word.sh
Enter a word
awDIU
Not Same
ubuntu@ubuntu:~/l4$ bash word.sh
Enter a word
DIU
Same
ubuntu@ubuntu:~/l4$

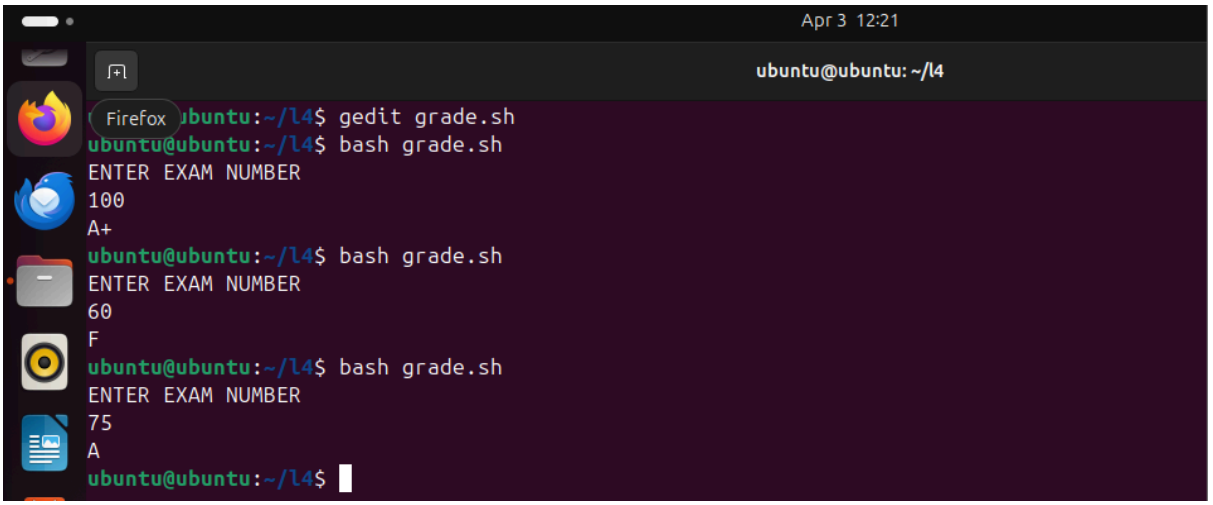

Open
word.sh
~/l4
Save
#!/bin/bash
echo "Enter a word"
read w
if ["$w" == "DIU"]
then
echo "Same"
else
echo "Not Same"
fi


Apr 3 12:10
ubuntu@ubuntu: ~/l4
ubuntu@ubuntu:~/l4$ gedit word.sh
ubuntu@ubuntu:~/l4$ bash word.sh
Enter a word
DIU
Same
ubuntu@ubuntu:~/l4$ bash word.sh
Enter a word
diu
Same
ubuntu@ubuntu:~/l4$


Open
*word.sh
~/l4
#!/bin/bash
echo "Enter a word"
read w
if ["$w" == "DIU"]
then
echo "Same"
elif ["$w" == "diu"]
then
echo "Same"
else
echo "Not Same"
fi


Apr 3 12:16
ubuntu@ubuntu: ~/l4
ubuntu@ubuntu:~/l4$ gedit division.sh
ubuntu@ubuntu:~/l4$ bash division.sh
Enter Two Numbers:
12 3
4.000
ubuntu@ubuntu:~/l4$ bash division.sh
Enter Two Numbers:
12 25
.480
ubuntu@ubuntu:~/l4$ bash division.sh
Enter Two Numbers:
23 -5
DIV not possible
ubuntu@ubuntu:~/l4$


Open
division.sh
~/l4
Save
1 #!/bin/bash
2
3 echo "Enter Two Numbers:"
4 read a b
5
6 if (($b <= 0))
7 then
8 echo "DIV not possible"
9 else
10 r=$(echo "scale=3; $a/$b" | bc)
11 echo "$r"
12 fi
13


Apr 3 12:21
ubuntu@ubuntu: ~/l4
Firefox
ubuntu@ubuntu:~/l4$ gedit grade.sh
ubuntu@ubuntu:~/l4$ bash grade.sh
ENTER EXAM NUMBER
100
A+
ubuntu@ubuntu:~/l4$ bash grade.sh
ENTER EXAM NUMBER
60
F
ubuntu@ubuntu:~/l4$ bash grade.sh
ENTER EXAM NUMBER
75
A
ubuntu@ubuntu:~/l4$

Open | grade.sh ~/l4 | Save

```
#!/bin/bash

echo "ENTER EXAM NUMBER"
read n

if (( n>=80))
then
echo "A+"
elif (( n>=70))
then
echo "A"
else
echo "F"
fi
```

LAB 05:

~~Lab 5~~

- **Summation and Parity:** Take user input, then calculate the cumulative sum for each number up to n and check if that sum is even or odd.
  - *Example:*
    n=5
    1=1,odd
    2=3,odd
    3=6,even
    4=10,even
    5=15,odd

Apr 3 12:32

ubuntu@ubuntu: ~/l5

```
ubuntu@ubuntu:~/l5$ gedit loop.sh
ubuntu@ubuntu:~/l5$ bash loop.sh
Enter a number
1=1, Odd
2=3, Odd
3=6, Even
4=10, Even
5=15, Odd
6=21, Odd
7=28, Even
ubuntu@ubuntu:~/l5$
```

Thunderbird Mail

Open | loop.sh ~/l5 | Save

```
#!/bin/bash

echo "Enter a number"
read n
sum=0

for (( i=1; i<=n; i++))
do
sum=$((sum+i))
if (( sum%2 == 0 ))
then
echo "$i=$sum, Even"
else
echo "$i=$sum, Odd"
fi
done
```

LAB06:

Lab 6

- **Security:** Password generator using OpenSSL.
- **Logic Tasks:**
  - Reverse a number and check if the reversed number is the same as the original (Palindrome check).
  - Input a 3-digit number and find the sum of its digits.
- **Functions:**
  - Find the factorial of a number using a function call.
  - Add two numbers using a function call.

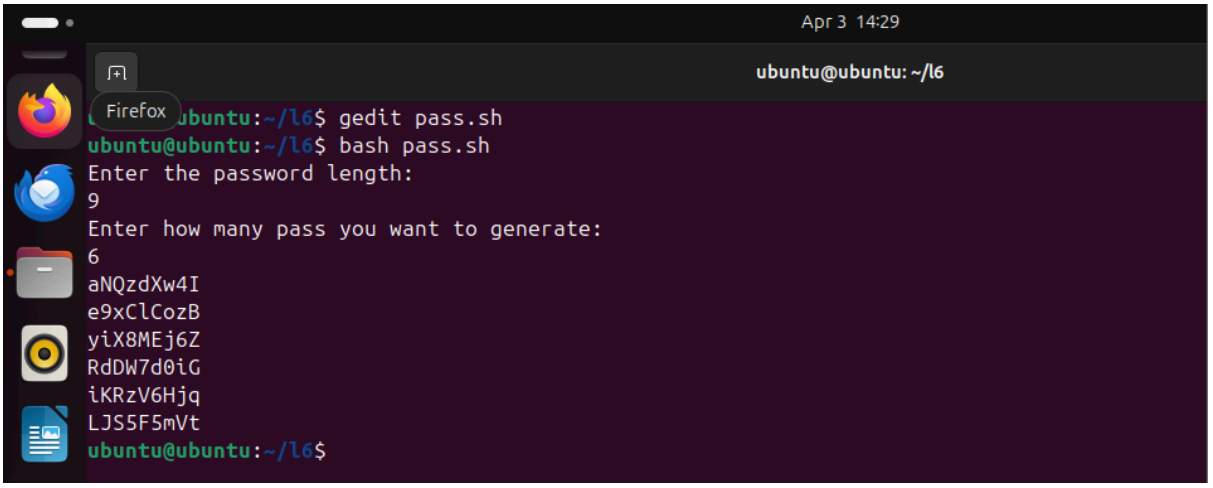
Apr 3 14:24

ubuntu@ubuntu: ~/l6

```
ubuntu@ubuntu:~$ mkdir l6
ubuntu@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  a  l2  l3  l4
ubuntu@ubuntu:~$ cd l6
ubuntu@ubuntu:~/l6$ gedit pass.sh
ubuntu@ubuntu:~/l6$ bash pass.sh
Enter the password length:
8
ztdmwV4K
ubuntu@ubuntu:~/l6$ bash pass.sh
Enter the password length:
5
4b4hx
ubuntu@ubuntu:~/l6$ bash pass.sh
Enter the password length:
15
GNAPIeG8iiIAhAx
ubuntu@ubuntu:~/l6$
```

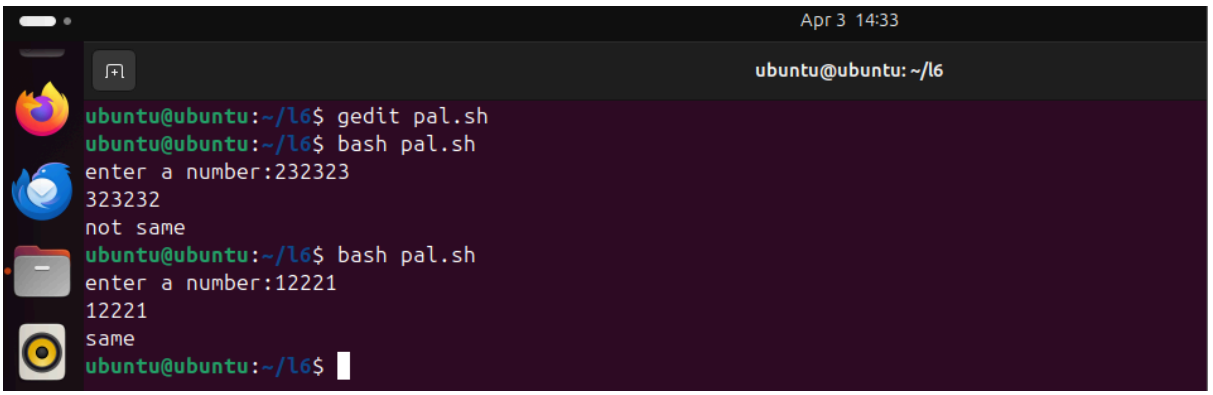
Open pass.sh ~/l6 Save

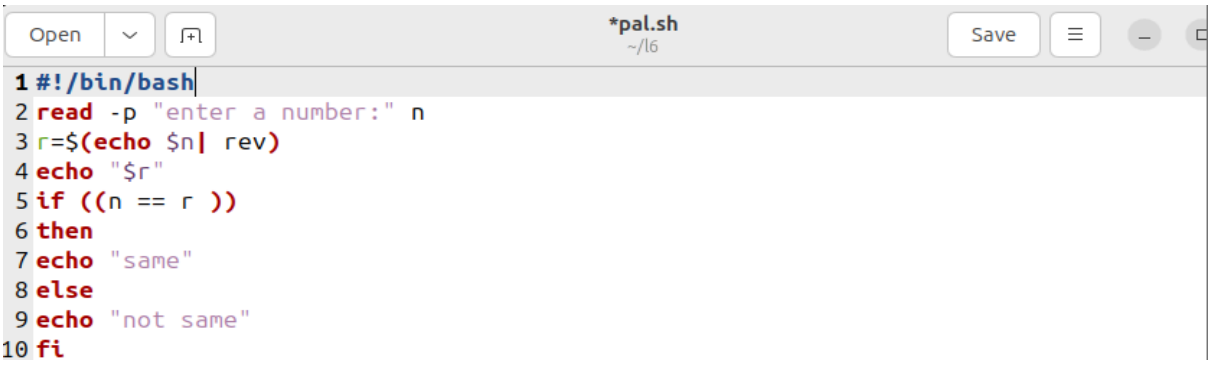
```
1 #!/bin/bash
2
3 echo "Enter the password length:"
4 read n
5
6 if [[ "$n" =~ ^[0-9]+$ ]]
7 then
8 openssl rand -base64 48 | head -c "$n"
9 echo
10 else
11 echo "enter valid length"
12 fi
```

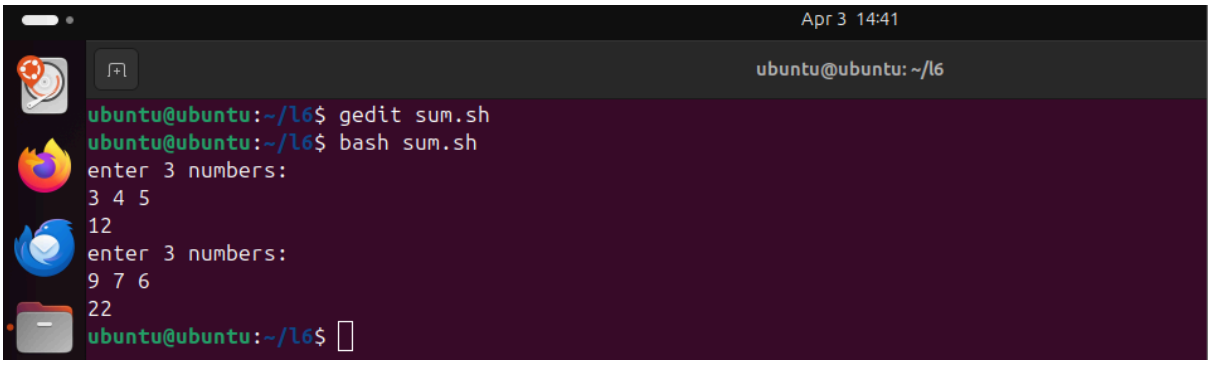


Open pass.sh ~/l6 Save

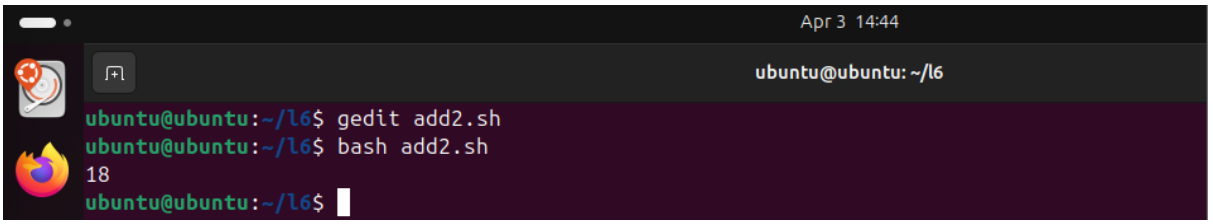
```
1 #!/bin/bash
2
3 echo "Enter the password length:"
4 read n
5 echo "Enter how many pass you want to generate:"
6 read a
7 if [[ "$n" =~ ^[0-9]+$ ]]
8 then
9 for (( i=1; i<=a; i++))
10 do
11 openssl rand -base64 48 | head -c "$n"
12 echo
13 done
14 else
15 echo "enter valid length"
16 fi
```


Apr 3 14:33
ubuntu@ubuntu: ~/l6
ubuntu@ubuntu:~/l6$ gedit pal.sh
ubuntu@ubuntu:~/l6$ bash pal.sh
enter a number:232323
323232
not same
ubuntu@ubuntu:~/l6$ bash pal.sh
enter a number:12221
12221
same
ubuntu@ubuntu:~/l6$


Open
*pal.sh
~/l6
Save
#!/bin/bash
read -p "enter a number:" n
r=$(echo $n| rev)
echo "$r"
if ((n == r))
then
echo "same"
else
echo "not same"
fi


Apr 3 14:41
ubuntu@ubuntu: ~/l6
ubuntu@ubuntu:~/l6$ gedit sum.sh
ubuntu@ubuntu:~/l6$ bash sum.sh
enter 3 numbers:
3 4 5
12
enter 3 numbers:
9 7 6
22
ubuntu@ubuntu:~/l6$

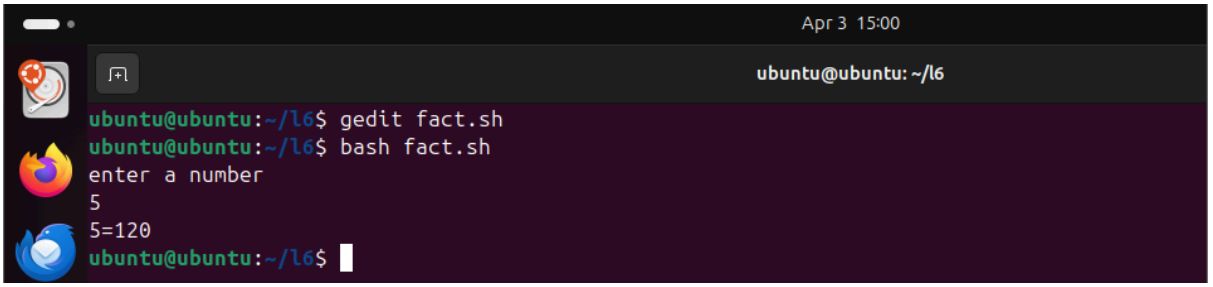

Open
sum.sh
~/l6
Save
#!/bin/bash
sum(){
echo "enter 3 numbers: "
read a b c
add=$((a+b+c))
echo "$add"
}
sum
sum


Apr 3 14:44
ubuntu@ubuntu: ~/l6
ubuntu@ubuntu:~/l6$ gedit add2.sh
ubuntu@ubuntu:~/l6$ bash add2.sh
18
ubuntu@ubuntu:~/l6$

Open | add2.sh ~/l6 | Save

```
#!/bin/bash
sum(){
add=$(( $1+$2+$3 ))
echo "$add"
}

sum 5 6 7
```



Open | *fact.sh ~/l6 | Save

```
#!/bin/bash
factorial (){
local n=$1
if (( n==1 ))
then
echo 1
else
prev=$(factorial $((n-1)))
r=$((n*prev))
echo "$r"
fi
}

echo "enter a number"
read n
fact=$(factorial $n)
echo "$n=$fact"
```